The effects of the dispersal ability in the diversification dynamics of Emberizoidea (Aves: Passeriformes)

Running title: Dispersal in speciation dynamics for Emberizoidea.

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The data underlying this article are available in FigShare at https://doi.org/10.6084/m9.figshare.29148893.v1 and can be freely accessed.

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We declare we have no conflicts of interest.

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Abstract

Understanding how ecological and evolutionary forces shape biodiversity is a core pursuit in macroevolutionary research. Species' dispersal ability significantly impacts their colonization chances and geographic isolation, profoundly influencing species ecology and evolution. Our study delved into the relationship between dispersal ability and speciation rates within Emberizoidea, a diverse group of passerine birds. We used the Hand-Wing Index (HWI) as a morphological proxy for dispersal ability and phylogenetic data for 749 species of Emberizoidea (~90% of all recognized species for the clade) using trait-dependent diversification models, including continuous-state (QuaSSE) and hidden-states (SecSSE) models. We also applied the Correlated Speciation and Trait Rates Simulation (Cor-STRATES) framework to assess correlation between HWI evolution and speciation rates. Our findings revealed that speciation in Emberizoidea was not significantly influenced by dispersal ability as estimated by HWI. While QuaSSE models suggested a peak in speciation rates at intermediate HWI values, the best-fit SecSSE model indicated that speciation was driven by unmeasured trait(s) rather than HWI. Moreover, HWI evolution and speciation rates were not correlated, suggesting that wing shape evolution is unrelated to cladogenetic events in Emberizoidea's evolutionary history. Our study challenges the notion that dispersal ability and wing shape drive diversification in birds, particularly in Emberizoidea.

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Introduction

The importance of the interactions between ecological and evolutionary processes in the origin of biodiversity is still one of the central questions in macroevolution (Imfeld & Barker, 2022). In fact, it is thought that the relationship between ecologically important traits and the rate at which new species arise explains major patterns of species distribution and diversity (Crouch & Ricklefs, 2019; Cooney & Thomas, 2021). Dispersal ability is a trait that directly influences the colonization likelihood of populations and the rate at which they develop geographical isolation from each other, promoting faster species divergence (Kennedy *et al.*, 2016). Accordingly, dispersal ability is regarded as significant trait with consequences for the ecology and evolution of species (Burges *et al.,* 2016; Claramunt, 2021, Arango et al. 2022).

The significance of dispersal ability for species generation is such that increases in diversity over time have frequently been associated to access of new regions of geographic or ecological space (Benton, 2009). Indeed, the ability to move to these new spaces is the central idea of the geographic radiations (Simões et al., 2016). According to this idea, species that have the capability to spread out and establish themselves in new environments would experience relief from various challenges like parasites and competitors (Ricklefs, 2011). At the same time, such species would be exposed to new ecological opportunities (Simpson, 1953) and different landscapes and topography (Cracraft, 1985). These factors would facilitate rapid speciation and specialization (Belmaker et al., 2012). Additionally, at the metapopulation level, the ability to disperse plays a crucial role in determining how quickly new areas are colonized and how many new individuals join existing populations.

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This process helps regulate gene flow and can potentially lead to genetic isolation (Kokko & Lopez-Sepulcre, 2006) which is often considered a necessary condition for speciation (Mayr, 1942; Rabosky, 2016).

Traits with ecological and/or evolutionary relevance (such as dispersal ability), can influence the rate at which lineages diversify, with certain trait states or values being associated with higher or lower speciation rates (Jablonski, 2008; Herrera-Alsina, 2019). Beyond the effect of trait states, it has also been proposed that speciation rates may correlate with the evolutionary rates of the traits themselves, that is the speed at which traits change over time (Stanley, 1979; Reznick & Ricklefs, 2009; Shi et al., 2021). This idea is particularly relevant in the context of adaptive radiations, where rapid trait evolution can promote increased lineage splitting (Schluter, 2000). Disentangling these potential drivers of diversification thus requires the combined evaluation of trait variation as well as trait evolutionary rate and their effect on diversification dynamics.

In birds, dispersal ability is regarded as a crucial trait driving diversification, but the direction of this relationship remains unclear (Claramunt et al., 2012). While high dispersal ability is thought to be linked to higher reproductive isolation given that broadly distributed lineages are prone to be interrupted by barriers to dispersal (Mayr, 1963), it would also enable gene flow among populations, preventing genetic differentiation (Kisel & Barraclough, 2010). Conversely, low dispersal ability would limit gene flow among populations, aiding their reproductive isolation (Gavrilets, 2003). However, lineages with low dispersal abilities would maintain more genetic cohesion by not being able to geographically separate (Claramunt et al., 2012), slowing their divergence rates (Gavrilets

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& Vose, 2005). To reconcile this conflicting relationship between dispersal ability and speciation, an intermediate dispersal model has been proposed (Price & Wagner, 2004; Claramunt et al., 2012). This model considers that species with intermediate dispersal abilities have higher speciation rates by providing more opportunities of range expansion, but also of geographic isolation, avoiding genetic cohesion among populations and aiding genetic differentiation (Claramunt et al., 2012).

Dispersal ability is strongly influenced by flight efficiency, which in turn is closely linked to wing morphology (Sheard et al., 2020; Claramunt, 2021; Weeks et al., 2022). A widely used morphological proxy for flight efficiency, and by extension, dispersal ability, is the Hand-Wing Index (HWI), which is related to the wing's aspect ratio and represents its elongation by comparing the length of the hand-wing relative to total wing length (Claramunt et al., 2012; Arango et al., 2022). Higher HWI values generally correspond to more elongated wings, which reduce energy expenditure during flight and are associated with greater natal and adult dispersal distances (Claramunt, 2021). Conversely, species with lower HWI values tend to have less elongated wings, more suited to maneuverability in dense habitats but less efficient for long-distance movement. While morphological indices are imperfect proxies, HWI has been repeatedly validated against direct measures of dispersal (Claramunt 2021; Weeks et al., 2022; Bastidas-Urrutia et al., 2024), making it a suitable trait for macroevolutionary analyses of dispersal in birds.

Here, we examined the relationship between dispersal ability and speciation rates in the Emberizoidea superfamily, a large clade of passerine birds (c. 832 spp; Barker et al., 2004) whose species have inspired important ecological and evolutive questions, from Darwin's

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finches to MacArthur's wood warblers. The clade Emberizoidea has been regarded as part of an important adaptive radiation (Barker et al., 2013; Lefebvre et al, 2016). This clade occurs in almost all the New World and parts of the Old World, exhibiting a wide range of ecological, behavioral, and morphological diversity (Lowther, 1975; Holmes, 1990; Forshaw & Parkes, 1991; Curson, 1994), in addition to an evident variation in species diversity, with clades comprising around 400 species such as Thraupidae to monotypic clades such as Rhodinocichlidae (Barker et al., 2013; 2015). Their ubiquitous distribution in the New World, both in mainland and islands, as well as their colonization of the Old World and Galapagos are a testimony of this clade's dispersal abilities (Barker et al., 2015). The contrasting dispersal abilities and diversity within clades of Emberizoidea provides an ideal system to test the effects of dispersal ability on diversification dynamics.

We used the HWI to evaluate the potential influence of dispersal ability on speciation rates through two complementary approaches. First, we applied trait-dependent diversification models (QuaSSE and SecSSE) to test if speciation rates were influenced by HWI values. We included hidden state models to account for potential heterogeneity in speciation rates not captured by HWI, thus avoiding spurious associations between our focal trait and diversification. Second, we used the Cor-STRATES framework to examine if speciation rates were correlated with the evolutionary rate of HWI, thus capturing different aspects of trait-speciation dynamics. Finally, we performed a Bayesian Multilevel regression model to test whether the observed relationship between evolutionary rate of HWI and speciation rate was consistent across the families within Emberizoidea.

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Materials and Methods

Dispersal ability data

We used the Hand-Wing Index (HWI) as a morphological proxy for dispersal ability (Sheard et al., 2020; Claramunt et al., 2021; Arango et al., 2022; Weeks et al., 2022, Bastidas-Urrutia et al., 2024). We compiled a dataset comprising a total of 749 spp (~90% of the total species of Emberizoidea), based primarily on the wing measurements from Arango et al. (2022; 490 spp), which we then completed with additional species from the AVONET database (Tobias et al., 202; 259 spp).

HWI data in the Arango et al. (2022) dataset was measured following Claramunt et al. (2012): $HWI = 100(\frac{WL-SL}{WL})$, using the standard length of the closed wing (WL) and the distance from the carpal joint to the tip of the first secondary feather (SL). AVONET states that their calculation of HWI used Kipp's distance instead of the difference between WL and SL, resulting in: $HWI = 100(\frac{Kipp's\ distance}{WL})$. To validate the combination of measurements, we compared HWI values from both datasets using their shared species and found a strong correlation (n= 258; r= 0.83; Figure S1), confirming that the AVONET data were comparable and could be reliably integrated with our measurements for subsequent analyses.

Phylogenetic data

We used Barker et al. (2015) Maximum Credibility Clade (MCC) tree. This phylogeny is the most comprehensive species level phylogeny for Emberizoidea, with 791emberizoid

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species of an approximate ~832 (95%). This phylogeny was estimated using a planned supertree method. This required assembling time-calibrated tree posteriors previously generated on a planned backbone. See Barker et al. (2015) for details of phylogenetic reconstruction. We pruned the MCC phylogeny to correspond with the species with HWI data. Although other avian mega phylogenies with a broader taxonomic scope are available, they are less suitable for their use with the Emberizoidea clade because they either have unresolved relationships among the clades of interest within Emberizoidea, such as an outdated backbone for the clade separation of Passerellidae and Emberizidae, lack resolution at the species level relationships, especially within Parulidae, are synthetically constructed (e.g., Jetz et al., 2012) or were not specifically designed for this clade (e.g., Oliveros et al., 2019).

Dispersal ability and Speciation

To test the effect of dispersal ability on speciation, we used Quantitative State Speciation and Extinction (QuaSSE) models. This approach allows us to test and compare a board array of possible evolutionary hypotheses that would reflect the general trends affecting patterns of diversification (FitzJohn, 2010). We fitted models representing the potential relationships between dispersal ability (approximated by HWI) and speciation (i.e.: linear, sigmoidal, and hump-shaped functions), which we then tested against a constant model (this model indicates that the trait holds no relationship with speciation), performing a model comparison based on AIC and AICw . We tested a total of seven different scenarios of how dispersal ability might have affected speciation in Emberizoidea (table 1); these being: 1) a constant relationship (dispersal ability does not have an effect on speciation), 2) a linear relationship (an increase on dispersal ability results in either an increase or decrease

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of speciation), 3) a sigmoidal relationship (dispersal ability values have an inflexion point in where constant speciation increases or decreases), 4) a unimodal relationship (dispersal ability values have an optimum in which speciation maximizes), and 5-7) the same linear, sigmoidal, and unimodal models but incorporating a drift parameter, which allows for directional trends in trait evolution (Figure S2; table 1).

In these models, trait drift represents the directional tendency of a continuous trait to increase or decrease over time, which may be driven by selection or any other process that has directional tendency (FitzJohn, 2010). In this case, the drift parameter allows to test whether there was an evolutionary trend in HWI values across Emberizoidea. A negative drift estimate would indicate a tendency for HWI to decrease over time, producing less elongated wings, and potentially reducing dispersal ability. In contrast, positive drift would indicate a tendency of HWI to increase over time, producing more elongated wings, and potentially enhancing dispersal ability. Thus, incorporating a trait drift term enables the detection of scenarios where traits evolve directionally over time, due to selection or other processes, even towards values that may be associated with reduced diversification (Clauset & Erwin, 2008; FitzJohn, 2010).

While traditional SSE models are prone to elevated Type 1 error rates (Rabosky & Goldberg, 2015), more recent implementations that incorporate the effect of hidden or unmeasured traits have substantially improve their robustness (Herrera-Alsina, 2019). Therefore, to corroborate the effects of HWI on Emberizoidea speciation when considering potential unmeasured traits, we compared four SSE models that contemplate the possibility that diversification may not be related to the focal trait but rather to some other

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unknown/unmeasured trait. Given that these Hidden State Speciation Extinction models have not yet been developed for continuous traits, we used a model that considers discrete trait states, namely the Several examined and concealed State-dependent Speciation and Extinction (SecSSE) model (Herrera-Alsina et al., 2019). For this, we classified HWI into three categories: Low (6.4 – 18.1), Intermediate (18.1-25.8), and High (25.8-39.4), using the Natural breaks method. Natural breaks (Jenks) classification is a clustering technique that divides a dataset into a specified number of uniform attribute groups. It works by maximizing the variation between different groups while minimizing the variation within each group (Jenks, 1967). Using uniform discrete categories has been recently applied for SecSSE models (e.g., Hill et al., 2023) as it minimizes the known bias of SSE models caused by overrepresentation of one trait state and thus maximizes the power of the analysis (Davis et al., 2013).

We tested four scenarios using the SecSSE framework: (i) diversification rates are influenced by HWI (EDT1); (ii) diversification rates are not influenced by HWI but by concealed (unmeasured) traits (CDT1); (iii) diversification rates are influenced by HWI, which also has a directional evolutionary trend (i.e., wings become more or less elongated through time; EDT2); (iv) diversification rates are influenced by unmeasured traits that also have directional evolutionary trends (CDT2) and (v) diversification rates are constant across trait states (CR). We tested the fit of all models based on AIC and AICw.

Dispersal-ability evolution and Speciation

To test the correlation between speciation rates and the evolutionary rate of HWI as an approximation of dispersal ability, we applied the correlated speciation and trait rates

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simulation (Cor-STRATES) framework (Cooney & Thomas, 2021). This approach allowed us to compare the observed correlation between the speciation rate and the rate of trait evolution with a null set of correlations generated by simulations. To do so, we first used the Bayesian cladogenetic diversification rate shift (ClaDS) model (Maliet et al., 2019) to calculate speciation rates (λ_{ClaDS}). This calculation was performed using the PANDA (Maliet & Morlon, 2022) package for julia (Bezanson et al., 2017) taking the appropriate sampling fraction and subsequently processed with the help of the RPANDA (Morlon et al., 2016) package for R. ClaDS provides high-resolution, branch-specific estimates of speciation rates by modeling gradual, lineage-specific rate changes, offering a more realistic and statistically robust alternative to models that assume rare, abrupt diversification shifts (Maliet et al., 2019). Subsequently, we calculated the Pagel's lambda on the observed HWI data and used the values of λ to rescale the Emberizoidea pruned tree using the package "geiger" for R (Harmon et al., 2009). This rescaling is done to adjust for the phylogenetic signal in trait data, which helps accounting for potential non-phylogenetic variation in trait values and thus reduces the chance for spurious correlations between rates (Cooney & Thomas, 2021). To calculate the rates of trait evolution of our observed HWI data, we used the lambda rescaled tree and fitted a trait evolution model using the Bayesian Analysis of Macroevolutionary Mixture (BAMM) (Rabosky, 2014). We used the setBAMMpriors function to establish the evolutionary priors, and to calculate the per-branch estimates of evolutionary rate (β_{BAMM}) we used the 'getMeanBranchLengthTree' function from the R package BAMMtools 2.1.7 (Rabosky et al., 2014).

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We calculated the observed correlation of the per-lineage speciation rate with the estimates of trait evolutionary rate using Spearman's rank correlation (ρ). Spearman's rank correlation is used in this case given the non-normal distribution of both speciation and phenotypic evolutionary rates (Cooney & Thomas, 2021). Note that this analysis is not directly corrected for phylogeny, but instead the statistical significance of the observed correlation is assessed by contrasting it against a null distribution of correlation values derived from a null model in which the trait of interest evolves neutrally, hence intrinsically considering phylogenetic relatedness. For the calculation of the null correlations, we first fitted a Brownian Motion model on the observed HWI data using the rescaled phylogeny, acquiring an estimated value of diffusion rate (σ^2). Then, constructed 100 datasets with simulated trait data using the σ^2 value. Afterwards, we recalculated the per-branch trait evolution for each simulated dataset by using BAMM with the same approach as the observed rates of trait evolution and estimated the null correlations (ρNull) between each of the simulated rates of trait evolution with the observed lineage speciation rate. Finally, we computed a two-tailed *P* value for the observed correlation against the null correlations. This framework tests whether the observed correlation between speciation rate and trait evolutionary rate is stronger than expected under a phylogenetically structured, but neutral, evolutionary process.

Furthermore, to test whether the observed relationship between HWI evolutionary rate and speciation rate was consistent across the Emberizoidea superfamily, we performed a Multilevel regression model using the "brms" R package and default priors (Burkner, 2017). We used branch-specific estimates of HWI evolutionary rates, derived from the

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298 BAMM model of phenotypic evolution with a lambda-transformed tree, and branch-
299 specific speciation rates estimated using the ClaDS model for all branches in the
300 phylogeny. To account for the phylogenetic non-independence and capture potential
301 differences among subclades within Emberizoidea, we modeled taxonomic families, which
302 are themselves monophyletic, as a grouping factor. This approach allowed us to investigate
303 the effect of taxonomic hierarchy (family-specific effect) in the relationship between HWI
304 evolutionary rates and speciation rates across Emberizoidea. This model was fit using four
305 No-U-Turn Sampler (NUTS) sampling chains for 5,000 generations, discarding 20% of
306 each run as burn-ins.

307 Results

308 Dispersal ability and speciation

309 The best fitting QuaSSE model was the unimodal + drift relationship (χ^2 = 64.4, d.f. = 7;
310 Table 1; Figure 1). The unimodal + drift model suggests that the optimal HWI values are
311 around 24 and 26, while HWI itself shows a trend of decreasing over time (drift = -0.49).
312
313 In the SecSSE models, when not considering unmeasured traits (i.e., ETD models), the
314 higher speciation rates were still attributable to Intermediate HWI (0.289 sp/My) in
315 comparison to Low HWI (0.208 sp/My) and High HWI (0.198 sp/My). However, the best-
316 fitting SecSSE model (CTD2) implied that speciation rate variation could not be attributed
317 to HWI variation but to other, hidden/unmeasured traits (Table 2). Additionally, model
318 selection further supports an evolutionary trend towards less elongated wings in
319 Emberizoidea, with a higher transition rate to decreasing HWI (0.196) than to an increase

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320 (0.038). A sensitivity analysis using only AVONET data produced qualitatively similar
321 results (Tables S1-S2; Figure S3).
322

323 Dispersal ability evolution and speciation

324 Using the Cor-STRATES framework, we calculated the lineages' speciation rate (λ_{ClaDS};
325 min = 0.04 sp/my (log -3.02), max=5.18 sp/my (log 1.6), mean= 0.45 sp/my (log -1.17);
326 Figure 2a; Figure S4). Lineage-specific speciation rates estimated with BAMM as
327 suggested by Cooney & Thomas (2021), were strongly correlated to those calculated by
328 ClaDS, ensuring consistency across methods (Figure S5). Pagel's λ for the observed HWI
329 (dispersal ability) used to rescale the MCC phylogeny showed a high and significant
330 phylogenetic signal ($\lambda = 0.83$, $p < 0.01$). This rescaled phylogeny was used to calculate the
331 lineages' mean rate of trait evolution, where we found that the evolutionary rate of HWI
332 had two predominant rates (β_{BAMM} ~ 3.6 HWI/my (log 1.3) and ~ 1.5 HWI/my (log 0.4);
333 Figure S6). The low-rate value comes from two lineage-wide decelerations at the
334 Thraupidae family crown and the *Atlapetes-Melozone* common ancestor, as well as two
335 accelerations in the *Nesospiza-Melanodera* common ancestor and in *Tangara chilensis*
336 (β_{BAMM}; min = 0.95 HWI/my (log -0.051), max = 12.19 HWI/my (log 2.5), mean = 2.33
337 HWI/my (log 0.84); Figure 2b). Although, we detected a positive significant correlation
338 between speciation rates and HWI evolutionary rates using Spearman's rank correlation (ρ
339 $= 0.216$, $p < 0.01$; Figure 3), there was no significant difference between the Null
340 correlations (ρ_{NULL}; min = -0.54, max = 0.45, mean = -0.35) and the observed ρ with the
341 ρ_{NULL} distributions ($p = 0.27$, [-0.51:0.39, 95% confidence intervals (CI)], SES = 2.64;
342 Figure 4). Since the critical test in Cor-STRATES is whether observed correlations are

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different than what would be expected under neutral trait evolution, this finding suggests that the association between HWI evolutionary rate and speciation rate is not stronger than expected under neutral trait evolution, thus not supporting a biologically meaningful link between dispersal trait dynamics and diversification in Emberizoidea.

Using the Multilevel Regression model, we found that the estimated standard deviation (σ) of the relationship between the evolutionary rate of HWI and speciation rate varied substantially (Mean $\sigma = 0.56$ [0.31:0.99; 95% credible intervals]). Further exploration of the parameter estimations indicated that the families Thraupidae and Passerellidae have a strong negative effect on the association between the rates of trait evolution and speciation. In contrast, Emberizidae and Cardinallidae show a positive effect—although the 95% credible intervals overlap with zero (Figure S7).

Discussion

Our results suggest that the speciation in Emberizoidea is not driven by wing shape, described by the Hand-Wing Index (HWI) and considered as a proxy of dispersal ability. Accordingly, we can infer that the ability to enter new ecological and geographical spaces through dispersal did not have a noticeable effect in the clade's diversification dynamics (Claramunt et al., 2012; Arango et al., 2025). HWI itself appears to be decreasing over time, implying that more recent species in Emberizoidea may have lower dispersal ability, possibly due to selective pressures or trait drift along the history of the clade. Previous work has shown that HWI evolution in Emberizoidea follows an Ornstein-Uhlenbeck pattern (Arango et al., 2022), which is consistent with a tendency towards an adaptive

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optimum, though the underlying processes remain uncertain. Moreover, we found no evidence that the evolutionary rate of HWI is more strongly linked to speciation rates than what would be expected under a null model of neutral trait evolution (i.e., Brownian motion), suggesting that species divergence in Emberizoidea were not followed or caused by the evolutionary dynamics of HWI and thus dispersal ability.

While HWI can in principle promote speciation by enabling colonization of new environments through dispersal ability, where divergence would occur due to geographic isolation (Gavrilets & Vose, 2005), our analyses indicate that this role is limited in Emberizoidea. Our results provide evidence that speciation across Emberizoidea seems to be driven instead by other unmeasured traits, even when certain lineages with intermediate values of HWI may be associated with higher speciation rates. In fact, there is evidence for some Neotropical taxa within this superfamily such as *Geothlypis aeoquinoctalis* and *Emberizoides herbicola* where vicariance and dispersal ability appear to play a minor role in divergence, with lineage histories instead reflecting ecological dynamics of the habitats they inhabit (van Els et al., 2021). Similarly, low divergence across large geographic distances in taxa with poor dispersal ability (Bates et al., 2003) and the importance of the lineages' intrinsic ability to persist in the landscape (i.e., local adaptation), without invoking dispersal ability (Smith et al., 2014), point to ecological or behavioral mechanisms, rather than dispersal per se, as drivers of speciation. Although certain species within Emberizoidea, such as *Leistes militaris* and *Leistes superciliaris*, may have resulted from divergence through dispersal events associated with seasonal migratory straying, thus highlighting gene flow and isolation effects linked to migratory behavior (Gomez-Bahamón et al., 2020; van Els et al., 2021), other species such as the Song sparrow (*Melospiza*

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389 *melodia)* could be differentiating through ecological divergence rather than dispersal per se,
390 highlighting the role of adaptation to distinct environments (Patten, 2010). Together this
391 evidence suggests that dispersal ability may facilitate population separation, but other traits
392 such as ecological specialization or species behavior likely interact with it to shape lineage
393 divergence. For example, for Icteriade, Passerellidae and Thraupidae, beak shape closely
394 related to dietary specialization has been correlated with their speciation rates (Conway &
395 Olsen, 2019). Similarly, traits related to plumage and song evolution have been suggested
396 as evolutionary drivers of Emberizoidea clades (Price & Eaton, 2014; Price-Waldman,
397 2019; Cicero et al., 2020).
398
399 This limited role of HWI in driving diversification in Emberizoidea contrasts with patterns
400 observed in another major avian clades. In furnariids, for example, low dispersal ability is
401 closely related to isolation and diversification (Claramunt, 2012). Emberizoidea, by
402 contrast, spans a wider geographic range and includes a higher number of migratory species
403 across several biographic realms (i.e., Neotropics, Afrotropic, Nearctic, and Palearctic;
404 Barker et al., 2015), yet dispersal ability, as estimated by HWI, has only a minor influence
405 on the clade's diversification. Instead, our best-fitting model indicates that heterogeneity in
406 speciation rates is not explained by HWI. While we cannot determine the specific causes of
407 this heterogeneity in our analysis, it may potentially involve ecological specializations,
408 behavioral syndromes, life-history strategies, or other unmeasured traits, as drivers of
409 lineage diversification. Recently, similar results were found when considering migratory
410 behavior for Emberizoidea and the suboscines, with no consistent relationship between
411 migration and speciation at higher taxonomic levels (Calabrese et al., 2025).
412

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The observed pattern of HWI decrease overtime reinforce this interpretation. Dispersal ability in birds tend to be reduced when species adapt to specialized foraging strategies (Forstmeier & Keßler, 2001; Jocque et al., 2010; Weeks et al., 2022) or predator avoidance behaviors (Swaddle & Lockwood, 2003). For example, in starling (*Sturnus vulgaris*) populations, HWI decreased by ~3.8% within 120 years after the colonization of new habitats (Bitton & Graham, 2014). In Emberizoidea diversification and functional evolution are considerably faster than expected by birth-death processes (Imfeld & Barker, 2022), consistent with ecological adaptation fostering speciation. Neotropical Emberizoids, in particular, display remarkable ecological versatility, exploiting diverse dietary and habitat niches, fostering speciation (Lefebvre et al., 2016, Vinciguerra & Burns, 2021; Conway & Olsen, 2019). Some appear to be limiting their species by habitat preference (D'Horta, 2009) or other behavioral traits such as feeding specialization (Phillimore et al., 2006) rather than geographical barriers like rivers or landscape processes (Bates et al., 2003; van Els et al., 2021; Naka et al., 2022).

Clade-specific patterns further highlight the complexity of the relationship of HWI with speciation. In Thraupidae (Figure S8), a clade mostly confined to the Neotropics (Winkler et al., 2020), we observed deceleration in HWI evolution, which together with their intense ecological adaptation through feeding specialization and innovations (Lefebvre et al., 2016; Vinciguerra & Burns, 2021), could suggest that specialization may reduce the relevance of dispersal ability causing evolutionary stasis in HWI (Eldredge et al., 2005). Yet, the importance of dispersal ability may re-emerge in island taxa in certain *Nesospiza* and *Melanodera* migrant species (Del Hoyo, 2020), where gene flow and isolation pressures would recover the usefulness for dispersal ability, breaking its evolutionary stasis (Eldredge

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et al., 2005). In the *Atlapetes-Melozone* lineage (Figure S9), HWI deceleration coincides with the *Atlapetes* speciation rate acceleration, though it is unclear whether this relationship is indicative of exploitation of ecological opportunities during the Andean uplift (Weir, 2006) or a reflection of Pleistocene isolation and drift (García-Moreno & Fjeldså, 2000; Sánchez-Gonzáles et al., 2014). These examples suggest that the role of dispersal ability in diversification may vary substantially across taxonomic scales and evolutionary history of particular clades, helping to explain why its overall relationship with diversification has remained elusive (Phillimore et al., 2006; Claramunt et al., 2012; Calabrese et al., 2025).

Comparisons with other taxa further illustrate this point. In furnariids high dispersal ability inhibits speciation, (Claramunt et al., 2012); in Corvids wing morphology is linked to both range size and diversification potential, although the strength and direction of these associations vary geographically (Kennedy et al., 2016). Outside birds, the link between dispersal and diversification is equally context-dependent, shaped by habitat structure and life-history traits (Burgess et al., 2016; Jablonski, 2008). Thus, the impact of dispersal ability on speciation may be strongly shaped by clade-specific ecological and evolutionary contexts, and broad generalizations may obscure this important variation. Altogether, these findings highlight the need to evaluate different proxies of dispersal ability and their influence on diversification at different phylogenetic scales.

In sum, our results challenge the assumption that dispersal ability, as estimated from HWI, is a major driver of speciation in Emberizoidea. Instead, our best-fitting SecSSE model indicates that heterogeneity in speciation rates is not explained by HWI. While we cannot determine the specific causes of this heterogeneity, previous work in Emberizoidea points

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to ecological factors (e.g., diet, song, plumage) as likely candidates (Price-Waldman, 2019; Conway et al., 2019; Cicero et al., 2020; Vinciguerra & Burns 2021). The observed decrease of HWI over time and the lack of correlation between its evolutionary rate and speciation further supports this view, suggesting that other traits more related to specialization would have driven diversification (Salisbury et al., 2012; Lefebvre et al., 2016; Vinciguerra & Burns, 2021; Conway & Olsen, 2019) regardless of dispersal events (Arango et al., 2025). Together, these findings highlight the need to explore traits beyond dispersal ability, such as feeding and behavioral specialization, and to consider temporal dynamics (e.g., through stratified state dependent diversification models) to better understand the drivers of Emberizoidea diversification.

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Table 1: QuaSSE Model Selection.

MODEL	LNLIK	K	AIC	ΔAIC	AICw
CONSTANT	-3068.1	3	6142.2	26.3	0
LINEAR	-3066.7	4	6141.4	25.5	0
LINEAR + DRIFT	-3066.7	5	6141.4	25.5	0
SIGMOIDAL	-3062.2	6	6138.4	22.5	0
SIGMOIDAL + DRIFT	-3061.6	7	6137.2	21.3	0
UNIMODAL	-3055.5	6	6123	7.1	0.03
UNIMODAL + DRIFT	**-3051**	**7**	**6115.5**	**0**	**0.97**

Table 2: SecSSE Model selection.

MODEL	LNLIK	K	AIC	ΔAIC	AICw
ETD1	-2482.89	6	4977.79	140.2	0
CTD1	-2447.71	6	4907.43	69.84	0
ETD2	-2477.78	6	4967.56	129.98	0
CTD2*	**-2412.79**	**6**	**4837.58**	**0**	**1**
CR	-2490.9	3	4987.8	150.21	0

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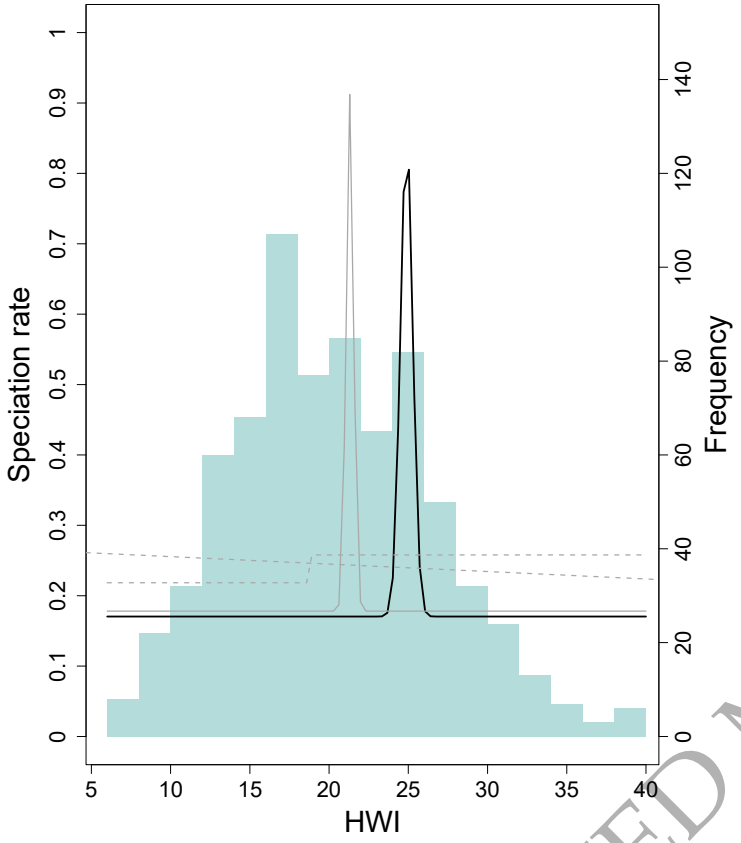

Figure 1: Relationships between dispersal ability and speciation rates in the Emberizoidea clade inferred from QuaSSE models. The histogram (y-axis= number of species per HWI bin) shows the distribution of HWI values across the clade. The gray line represents the unimodal model, and the black line represents the best fitting unimodal + drift model. Dashed lines represent the other models.

Alt text: A dual-axis plot showing the relationship between hand-wing index (HWI) and speciation rates in Emberizoidea. The x-axis shows HWI values. The left y-axis shows speciation rates, while the right y-axis shows number of species. A histogram displays the distribution of HWI values across species, with peak frequency around HWI of 20-25. Overlaid lines show different QuaSSE model fits: a gray line for the unimodal model, a black line for the best-fitting unimodal plus drift model, and dashed lines for alternative models. The figure shows higher speciation rates at intermediate HWI values.

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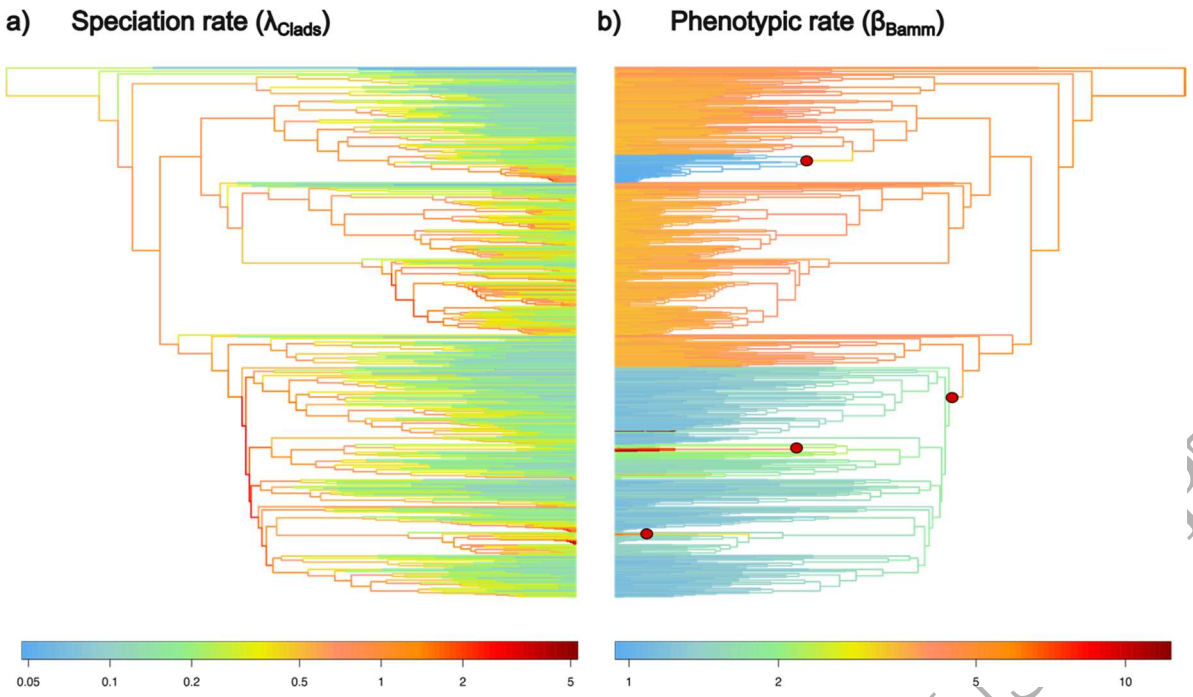

Figure 2: a) Speciation rate (λ_{ClaDS}) inferred from ClaDS using the time calibrated Emberizoidea phylogeny. b) HWI evolutionary rate (β_{BAMM}) inferred from BAMM using the lambda rescaled Emberizoidea phylogeny. Red circles are indicative of a rate shift for phenotypic rates, from top to bottom: *Atlapetes-Melozone* common ancestor, Thraupidae family crown, *Nesospiza-Melanodera* common ancestor, *Tangara chilensis*.
Alt Text: Two phylogenetic trees in which their branches represent Speciation rates calculated by ClaDS and Phenotypic evolutionary rates for HWI calculated by BAMM, illustrating variation in speciation and phenotypic evolution across clades. The phenotypic evolutionary rate tree also has circles indicating the nodes where rate shifts occurred.

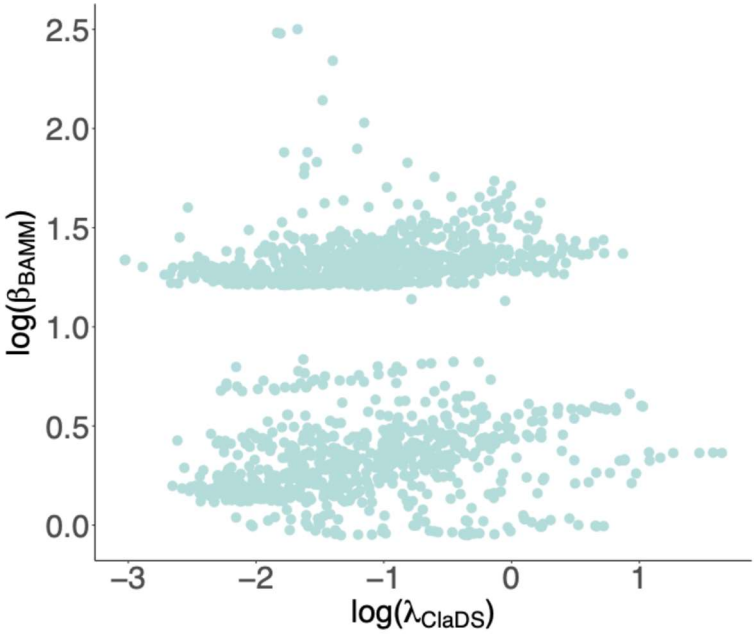

Figure 3: Correlation between lineage speciation rates (λ_{ClaDS}) and lineage HWI evolutionary rates (β_{BAMM}) of Emberizoidea.

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Alt Text: A scatterplot showing the correlation between speciation rate and phenotypic rate across clades. The x-axis represents log-transformed speciation rate (log $Lambda_{ClaDS}$), and the y-axis represents log-transformed phenotypic rate (log $Beta_{BAMM}$). Points are distributed along a slight positive trend.

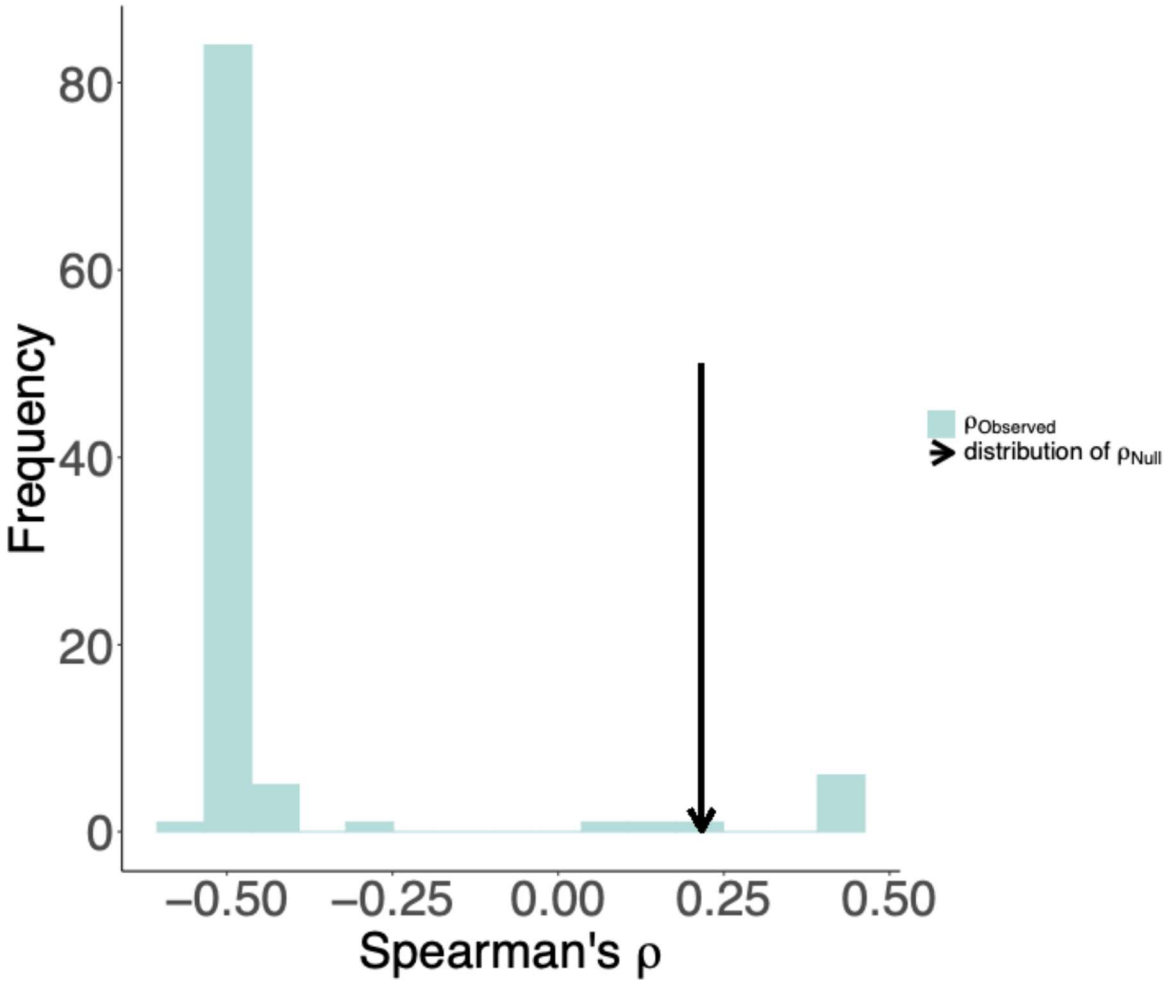

Figure 4: Correlation coefficient of the observed relationship between speciation rates and HWI evolutionary rates (arrow) in relation to a null distribution of correlation coefficients calculated from 100 simulated datasets.

Alt Text: A histogram showing the distribution of Spearman's rho correlation coefficients from a null model analysis. The histogram bars represent a null distribution of 100 simulated correlation coefficients between speciation rates and HWI evolutionary rates with ranges between -0.50 to 0.50. A vertical arrow labeled 'Observed rho' marks the observed correlation coefficient from the actual data, positioned within the null distribution.

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